Yunni Qu

quyunni@cs.unc.edu | quyunniii.github.io/

Education

University of North Carolina at Chapel Hill Aug. 2023 – present
Ph.D. in Computer Science (enroute M.S. recieved in 2025) Chapel Hill, NC

- Expected Graduation: Summer 2027
- Thesis Title (tentative): Towards Inference-Time Reliability & Acquisition Utility
- PhD Advisor: Junier Oliva
- **Methods**: Machine Learning - Longitudinal Active Feature Acquisition; Machine Learning - Out-of-distribution Generalization
- **Applications**: Health Care Prognosis (longitudinal); Psychological Assessment (longitudinal); Virtual Screening in Drug Discovery (highly imbalanced)

University of Toronto Sep. 2019 – Jun. 2023
Honours B.S. in Computer Science, Data Science with High Distinction Toronto, ON, CA

- **Coursework**: Data Structures, Algorithms, Databases, Artificial Intelligence, Machine Learning
- **Research**: Neural Network Uncertainty, Medical Image Segmentation and Classification

Peer-Reviewed Publications

- [5] Dinh, D.*, Chen, B.*, **Qu, Y.**, Niethammer, M., Oliva, J. 2026. **Contrastive Nonmyopic Objective Cost-Tradeoff Acquisition for Longitudinal Data.** 2026. Advances in Neural Information Processing Systems (NeurIPS) **[main track; acceptance rate: 25.7%]**

- [4] **Qu, Y.**, Dinh, D., King, G., Ringwald, W., Kok, B., Gates, K., Wright, A., and Oliva, J. 2026. **Relaxed Efficient Acquisition of Context and Temporal Features.** Proceedings of the 17th ACM International Conference on Bioinformatics, Computational Biology and Health Informatics (ACM BCB). Association for Computing Machinery, New York, NY, USA, Article 15, 110. https://doi.org/10.1145/3807503.3819475 **[long oral; acceptance rate: 19.9%]**

- [3] **Qu, Y.**, Vaduri, B., Jatoth K., Wellnitz, J., Dinh, D., Veenbaas, S., Chapman, J., Tropsha, A., and Oliva, J. 2026. **Reliable OOD Virtual Screening with Extrapolatory Pseudo-Label Matching.** Proceedings of the 17th ACM International Conference on Bioinformatics, Computational Biology and Health Informatics (ACM BCB). Association for Computing Machinery, New York, NY, USA, Article 37, 110. https://doi.org/10.1145/3807503.3819476 **[long oral; acceptance rate: 19.9%]**

- [2] Xing, E., Liu, L., Xing, X., **Qu, Y.**, Jacobs, N., & Liang, G. (2022, August). "**Neural Network Decision-Making Criteria Consistency Analysis via Inputs Sensitivity**," 2022 26th International Conference on Pattern Recognition (ICPR), Montreal, QC, Canada, 2022, pp. 2328-2334, doi: 10.1109/ICPR56361.2022.9956394.

- [1] **Qu, Y.**, Yan, D., Xing, E., Zheng, F., Zhang, J., Liu, L., & Liang, G. (2022, July). "**Beware the Black-Box of Medical Image Generation: an Uncertainty Analysis by the Learned Feature Space**," 2022 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), Glasgow, Scotland, United Kingdom, 2022, pp. 3849-3853, doi: 10.1109/EMBC48229.2022.9871921.

Preprints

- [preprint] **Qu, Y.***, Kok, B.*, Ringwald, W., King, G., Wright, A., Gates, K., and Oliva, J. **Interpretable Active Feature Acquisition for Cost-Efficient Temporal Prediction with Reduced Participant Burden.** https://arxiv.org/abs/2610.07452 2026.

Research Experience

University of North Carolina at Chapel Hill Aug. 2023 – Present
Graduate Research Assistant Chapel Hill, NC

- Research on longitudinal active feature acquisition and out-of-distribution methods for healthcare, drug discovery, clinical psychology.

University of Toronto Sep. 2022 – May 2023
Research Assistant Toronto, ON, CA

- Predict the environmental impact of future annual conferences energy use by international and domestic flights of attendee to international conferences.

Princess Margaret Cancer Centre May 2022 – Aug 2022
Research Intern Toronto, ON, CA

- Research on testing the robustness of cervical cancer MR image segmentation models (nn-UNet) on real hospital imaging data, using conventional dice metrics and state-of-the-art robustness metrics.

Texas A&M University Aug. 2021 – May 2022
Research Student Remote

- Research on uncertainty analysis of neural network models for medical image generation from low resolution to high resolution.

University of Toronto May 2021 – Aug. 2021
Research Assistant Toronto, ON, CA

- Analyzed experimental results and reported findings for medical image analysis research.

Teaching Experience

University of North Carolina at Chapel Hill Aug. 2026
Guest Lecturer Chapel Hill, NC

- Delivered PhD-level lectures on probability and optimization for COMP755 (Graduate-level Machine Learning).

University of North Carolina-PRIMES program May 2025 – Jul. 2025
Instructor Chapel Hill, NC

- Developed curriculum and lectured a machine learning short course for an residential summer research program in medical imaging and Alzheimer's disease.

Academic Service

Conference Reviewer:
AAAI Conference on Artificial Intelligence 2027
AAAI Conference on Artificial Intelligence 2026
International Conference on Learning Representations (ICLR) 2025

Honors

Dean's List Scholar 2023, 2022, 2021
ASA Data Fest Best Visualization 2022